Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.

Criteria assessment

This task sheet assesses Criteria C5 through C7. Each problem assesses the criterion indicated in its title, and each problem is worth 2 points. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers that do not include the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion.

A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C5 Applies the Minimax Regret criterion by constructing an opportunity-loss table and selecting the alternative with the smallest maximum regret.

C6 Calculates EMV using given probabilities and selects the alternative with the highest expected payoff.

C7 Uses subjective probabilities and sensitivity analysis to identify threshold probabilities and the preferred alternative over each feasible range.

1 Artisan Apiary Expansion [C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits (in thousands of dollars) depend on the honey season.

Alternative	Excellent	Average	Poor
Large	76	34	-14
Medium	58	49	23
Small	38	43	35

Use the Minimax Regret criterion to determine which plan should be selected. Show the best payoff in each state, the regret table, and each alternative's maximum regret. **[2]**

C5 — Solution

The best payoff in each state is

$$b(S_1) = \max\{76, 58, 38\} = 76, \quad b(S_2) = \max\{34, 49, 43\} = 49, \quad b(S_3) = \max\{-14, 23, 35\} = 35.$$

Using $r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$ gives

	S_1	S_2	S_3	Maximum regret
A_1	$76 - 76 = 0$	$49 - 34 = 15$	$35 - (-14) = 49$	49
A_2	$76 - 58 = 18$	$49 - 49 = 0$	$35 - 23 = 12$	18
A_3	$76 - 38 = 38$	$49 - 43 = 6$	$35 - 35 = 0$	38

Thus $\min\{49, 18, 38\} = 18$, so the Minimax Regret criterion selects A_2, the medium expansion.



2 Regional Theater Capacity [C5-2]

A theater operator is choosing a seating-capacity plan. Seasonal profit (in thousands of dollars) is shown below; no reliable probabilities are available.

Alternative	High	Normal	Low
Add two seating sections	108	40	-28
Add one seating section	82	61	14
Lease a nearby hall	59	52	34
Current seating	34	34	34

Determine the decision using Minimax Regret. **[2]**

C5 — Solution

The statewise best payoffs are $b(S_1) = 108$, $b(S_2) = 61$, and $b(S_3) = 34$.

	S_1	S_2	S_3	Maximum regret
A_1	$108 - 108 = 0$	$61 - 40 = 21$	$34 - (-28) = 62$	62
A_2	$108 - 82 = 26$	$61 - 61 = 0$	$34 - 14 = 20$	26
A_3	$108 - 59 = 49$	$61 - 52 = 9$	$34 - 34 = 0$	49
A_4	$108 - 34 = 74$	$61 - 34 = 27$	$34 - 34 = 0$	74

Since $\min\{62, 26, 49, 74\} = 26$, the recommended decision is A_2 : add one seating section.



3 Emergency Shelter Staffing [C5-2]

A relief organization must select an emergency-shelter staffing plan. Net community value (in thousands of dollars) depends on requests for shelter.

Alternative	Very high	High	Moderate	Low
Full	96	54	24	-24
Flexible	74	64	49	12
Limited	48	51	46	38

Use Minimax Regret to select a staffing plan. **[2]**

C5 — Solution

The best-payoff vector is

$$(b(S_1), b(S_2), b(S_3), b(S_4)) = (96, 64, 49, 38).$$

	S_1	S_2	S_3	S_4	Maximum regret
A_1	0	10	25	62	62
A_2	22	0	0	26	26
A_3	48	13	3	0	48

For example, $r(A_1, S_4) = 38 - (-24) = 62$ and $r(A_3, S_2) = 64 - 51 = 13$. The smallest maximum regret is $\min\{62, 26, 48\} = 26$, so select A_2, the flexible staffing plan.



4 Coffee Roastery Product Mix [C6-2]

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

	Strong	Stable	Weak
$P(S_j)$	0.45	0.35	0.20
Single-origin	86	44	-12
Balanced	64	58	22
Classic	42	47	37

Calculate every EMV and apply the Bayes decision criterion. **[2]**

C6 — Solution

First, $0.45 + 0.35 + 0.20 = 1$. Then

$$EMV(A_1) = (0.45)(86) + (0.35)(44) + (0.20)(-12) = 38.7 + 15.4 - 2.4 = 51.7,$$
$$EMV(A_2) = (0.45)(64) + (0.35)(58) + (0.20)(22) = 28.8 + 20.3 + 4.4 = 53.5,$$
$$EMV(A_3) = (0.45)(42) + (0.35)(47) + (0.20)(37) = 18.9 + 16.45 + 7.4 = 42.75.$$

The highest EMV is $\max\{51.7, 53.5, 42.75\} = 53.5$. Therefore, choose A_2, the balanced product mix, with expected profit \$53,500.



5 Aquarium Program Scale [C6-2]

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

	High	Expected	Low
$P(S_j)$	0.30	0.40	0.30
Major	98	38	-24
Standard	73	53	18
Compact	52	47	34
Regular tours only	27	27	27

Calculate the EMVs and recommend an alternative. **[2]**

C6 — Solution

The probabilities are valid because $0.30 + 0.40 + 0.30 = 1$.

$$EMV(A_1) = (0.30)(98) + (0.40)(38) + (0.30)(-24) = 29.4 + 15.2 - 7.2 = 37.40,$$
$$EMV(A_2) = (0.30)(73) + (0.40)(53) + (0.30)(18) = 21.9 + 21.2 + 5.4 = 48.50,$$
$$EMV(A_3) = (0.30)(52) + (0.40)(47) + (0.30)(34) = 15.6 + 18.8 + 10.2 = 44.60,$$
$$EMV(A_4) = (0.30)(27) + (0.40)(27) + (0.30)(27) = 27.00.$$

Because 48.50 is the highest EMV, the Bayes decision is A_2, the standard program, with expected payoff \$48,500.

6 Wind-Farm Maintenance Staffing [C6-2]

A wind-farm operator is choosing a maintenance staffing plan for next quarter. Payoffs are in thousands of dollars.

	Surge	Strong	Steady	Slow
$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring	105	64	24	-32
Moderate hiring	79	68	48	12
Current staff	53	55	51	42

Use EMV to determine the preferred staffing plan. **[2]**

C6 — Solution

The probability check is $0.20 + 0.30 + 0.35 + 0.15 = 1$.

$$EMV(A_1) = (0.20)(105) + (0.30)(64) + (0.35)(24) + (0.15)(-32) = 21 + 19.2 + 8.4 - 4.8 = 43.8,$$
$$EMV(A_2) = (0.20)(79) + (0.30)(68) + (0.35)(48) + (0.15)(12) = 15.8 + 20.4 + 16.8 + 1.8 = 54.8,$$
$$EMV(A_3) = (0.20)(53) + (0.30)(55) + (0.35)(51) + (0.15)(42) = 10.6 + 16.5 + 17.85 + 6.3 = 51.25.$$

Thus $\max\{43.8, 54.8, 51.25\} = 54.8$, so choose A_2, moderate hiring, with expected payoff \$54,800.

7 Concert Merchandise Order [C7-2]

A concert vendor must choose between a large merchandise order (A_1) and a small merchandise order (A_2). Payoffs (in thousands of dollars) depend on whether concert attendance is high (S_1) or low (S_2).

Alternative	High (S_1)	Low (S_2)
Large (A_1)	80	20
Small (A_2)	50	40

Let $P(S_1) = p$ and $P(S_2) = 1 - p$. Find the threshold probability and state the preferred alternative for every $0 \le p \le 1$. **[2]**

C7 — Solution

$$EMV(A_1) = 80p + 20(1-p) = 20 + 60p,$$
$$EMV(A_2) = 50p + 40(1-p) = 40 + 10p.$$

At the threshold, $20 + 60p = 40 + 10p$, so $50p = 20$ and $p = \frac{2}{5} = 0.4$. Since A_2 has the higher EMV at $p = 0$ and A_1 has the higher EMV at $p = 1$,

$$\begin{cases} A_2, & 0 \le p < \frac{2}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{2}{5}, \\ A_1, & \frac{2}{5} < p \le 1. \end{cases}$$



8 Meal-Kit Launch [C7-2]

A meal-kit company is comparing a bold launch (A_1) with a measured launch (A_2). Payoffs are in thousands of dollars.

Alternative	High	Moderate	Low
Bold (A_1)	60	30	0
Measured (A_2)	35	45	30

Suppose $P(S_1) = p$ and $P(S_2) = \frac{p}{3}$. Determine the feasible range of p, the threshold, and the preferred launch over the entire range. **[2]**

C7 — Solution

Because the three state probabilities must sum to one,

$$P(S_3) = 1 - p - \frac{p}{3} = 1 - \frac{4p}{3}.$$

Nonnegative probabilities require $p \ge 0$ and $1 - \frac{4p}{3} \ge 0$, so the feasible range is $0 \le p \le \frac{3}{4}$.

$$EMV(A_1) = 60p + 30\left(\frac{p}{3}\right) + 0\left(1 - \frac{4p}{3}\right) = 70p,$$
$$EMV(A_2) = 35p + 45\left(\frac{p}{3}\right) + 30\left(1 - \frac{4p}{3}\right) = 30 + 10p.$$

Equating the EMVs gives $70p = 30 + 10p$, hence

$$60p = 30 \quad \Rightarrow \quad p = \frac{1}{2}.$$

Therefore,

$$\begin{cases} A_2, & 0 \le p < \frac{1}{2}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{2}, \\ A_1, & \frac{1}{2} < p \le \frac{3}{4}. \end{cases}$$



9 Publishing-Platform Bid [C7-2]

A publishing firm must choose between a standard platform bid (A_1) and an experimental platform bid (A_2). Payoffs are in thousands of dollars.

Alternative	Immediate	Revision	Delayed
Standard (A_1)	30	70	70
Experimental (A_2)	50	50	30

Suppose $P(S_1) = 3p$ and $P(S_2) = p$. Find the feasible range, threshold probability, and preferred bid for each range of p. **[2]**

C7 — Solution

The remaining probability is

$$P(S_3) = 1 - 3p - p = 1 - 4p.$$

The probability conditions $p \ge 0$ and $1 - 4p \ge 0$ give the feasible range $0 \le p \le \frac{1}{4}$.

$$EMV(A_1) = 30(3p) + 70p + 70(1 - 4p) = 70 - 120p,$$
$$EMV(A_2) = 50(3p) + 50p + 30(1 - 4p) = 30 + 80p.$$

At indifference,

$$70 - 120p = 30 + 80p \quad \Rightarrow \quad 40 = 200p \quad \Rightarrow \quad p = \frac{1}{5} = 0.2.$$

Here A_1 is preferred below the threshold, while the faster-growing EMV of A_2 makes it preferred above the threshold:

$$\begin{cases} A_1, & 0 \le p < \frac{1}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{5}, \\ A_2, & \frac{1}{5} < p \le \frac{1}{4}. \end{cases}$$



10 Summarized Solutions

10.1 Artisan Apiary Expansion C5–2

The best-payoff vector is $(76, 49, 35)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	0	15	49	49
A_2	18	0	12	18
A_3	38	6	0	38

Thus $\min\{49, 18, 38\} = 18$, so select A_2, the medium expansion.

10.2 Regional Theater Capacity C5–2

The best-payoff vector is $(108, 61, 34)$. The regret table and row maxima are

	S_1	S_2	S_3	$\max r$
A_1	0	21	62	62
A_2	26	0	20	26
A_3	49	9	0	49
A_4	74	27	0	74

Thus $\min\{62, 26, 49, 74\} = 26$, so select A_2, add one seating section.

10.3 Emergency Shelter Staffing C5–2

The best-payoff vector is $(96, 64, 49, 38)$. The regret table and row maxima are

	S_1	S_2	S_3	S_4	$\max r$
A_1	0	10	25	62	62
A_2	22	0	0	26	26
A_3	48	13	3	0	48

Thus $\min\{62, 26, 48\} = 26$, so select A_2, flexible staffing.

10.4 Coffee Roastery Product Mix C6–2

Since $0.45 + 0.35 + 0.20 = 1$,

$$EMV(A_1) = (0.45)(86) + (0.35)(44) + (0.20)(-12) = 51.7,$$
$$EMV(A_2) = (0.45)(64) + (0.35)(58) + (0.20)(22) = 53.5,$$
$$EMV(A_3) = (0.45)(42) + (0.35)(47) + (0.20)(37) = 42.75.$$

The largest EMV is 53.5, so select A_2, the balanced product mix (\$53,500).

10.5 Aquarium Program Scale C6–2

$$EMV(A_1) = 37.40, \qquad EMV(A_2) = 48.50,$$
$$EMV(A_3) = 44.60, \qquad EMV(A_4) = 27.00.$$

Using probabilities $(0.30, 0.40, 0.30)$, the largest EMV is 48.50, so select A_2, the standard program (\$48,500).

10.6 Wind-Farm Maintenance Staffing C6–2

$$EMV(A_1) = (0.20)(105) + (0.30)(64) + (0.35)(24) + (0.15)(-32) = 43.8,$$
$$EMV(A_2) = (0.20)(79) + (0.30)(68) + (0.35)(48) + (0.15)(12) = 54.8,$$
$$EMV(A_3) = (0.20)(53) + (0.30)(55) + (0.35)(51) + (0.15)(42) = 51.25.$$

The largest EMV is 54.8, so select A_2, moderate hiring (\$54,800).

10.7 Concert Merchandise Order C7–2

$$EMV(A_1) = 20 + 60p, \qquad EMV(A_2) = 40 + 10p.$$

Indifference requires $20 + 60p = 40 + 10p$, so $p = \frac{2}{5} = 0.4$. Therefore

$$A_2 \ (0 \le p < \tfrac{2}{5}); \quad A_1 = A_2 \ (p = \tfrac{2}{5}); \quad A_1 \ (\tfrac{2}{5} < p \le 1).$$

10.8 Meal-Kit Launch C7–2

Here $P(S_3) = 1 - \frac{4p}{3}$, so $0 \le p \le \frac{3}{4}$. Also,

$$EMV(A_1) = 70p, \qquad EMV(A_2) = 30 + 10p.$$

Indifference requires $70p = 30 + 10p$, so $p = \frac{1}{2}$. Therefore

$$A_2 \ (0 \le p < \tfrac{1}{2}); \quad A_1 = A_2 \ (p = \tfrac{1}{2}); \quad A_1 \ (\tfrac{1}{2} < p \le \tfrac{3}{4}).$$

10.9 Publishing-Platform Bid C7–2

Here $P(S_3) = 1 - 4p$, so $0 \le p \le \frac{1}{4}$. Also,

$$EMV(A_1) = 70 - 120p, \qquad EMV(A_2) = 30 + 80p.$$

Indifference requires $70 - 120p = 30 + 80p$, so $p = \frac{1}{5} = 0.2$. Therefore

$$A_1 \ (0 \le p < \tfrac{1}{5}); \quad A_1 = A_2 \ (p = \tfrac{1}{5}); \quad A_2 \ (\tfrac{1}{5} < p \le \tfrac{1}{4}).$$